

# **ECEN 607: Advanced Analog Circuit Tech**

## **Lab 3: Op-Amp Design 1**

### **Characterization Lab Report**

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1. Simulate the circuit from the prelab and adjust the transistor sizes accordingly until all specifications are met. Provide necessary screen shots such that you clearly show the specs are met.  
 Transistor sizes screenshot:  
 Given spec:

**Table 3-1: Design specifications**

|                                         |                                                   |
|-----------------------------------------|---------------------------------------------------|
| $V_{DD}-V_{SS}$                         | 1.8V                                              |
| $A_{v0}$                                | > 40 dB                                           |
| GBW                                     | > 2 MHz                                           |
| PM                                      | > 45°                                             |
| Output Swing                            | > 1 V                                             |
| $C_L$                                   | 20 pF                                             |
| $R_L$                                   | 10k $\Omega$                                      |
| Input referred noise<br>(10 Hz – 2 MHz) | < 30 $\mu$ Vrms<br>(ignore the noise in this lab) |

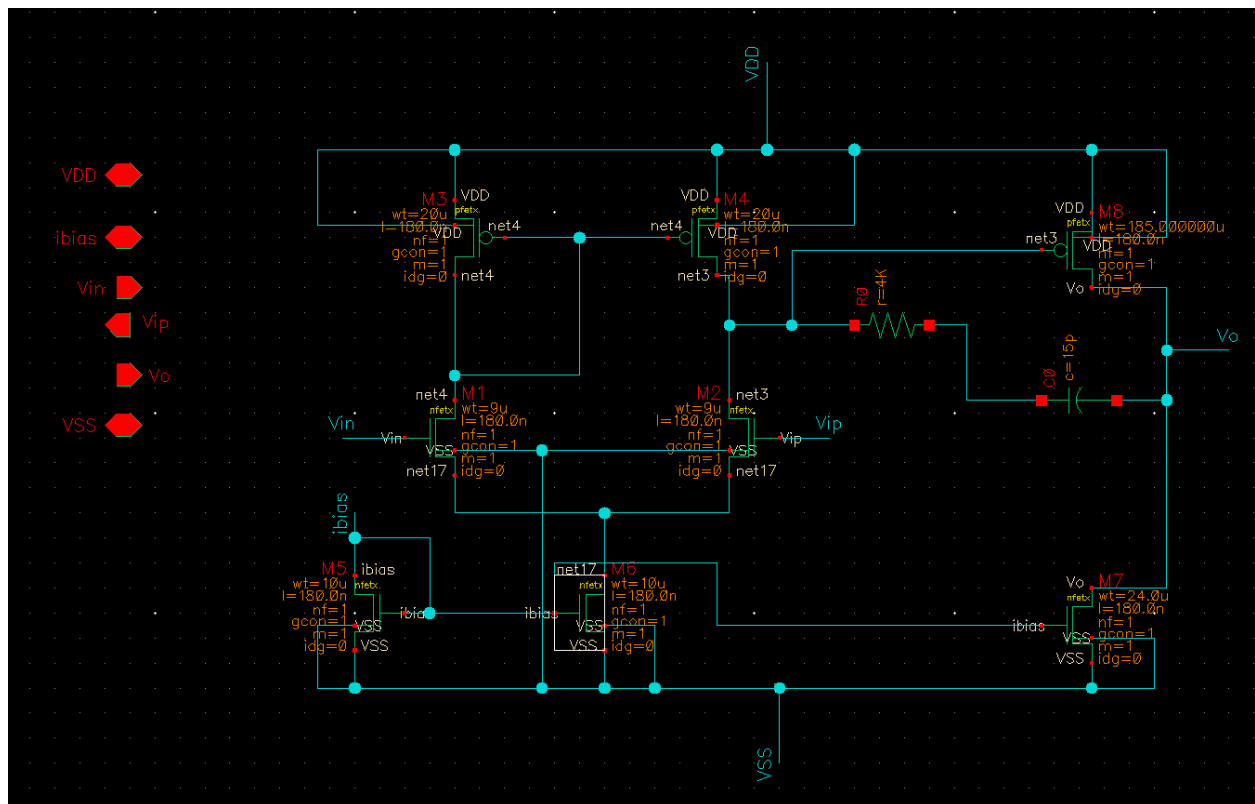
Obtained Spec:

|                      |          |
|----------------------|----------|
| Input referred Noise | 61.04 u  |
| CL                   | 20p      |
| RL                   | 10k      |
| PM                   | 83.76    |
| GBW                  | 63.47MHz |
| OUTPUT SWING         | 1.0412V  |
| GAIN(AV0)            | 41.737dB |
| VDD-VSS              | 1.8      |
| ID                   | 90u      |
| RC                   | 4k       |
| CC                   | 15p      |

Table showing the transistor sizing:

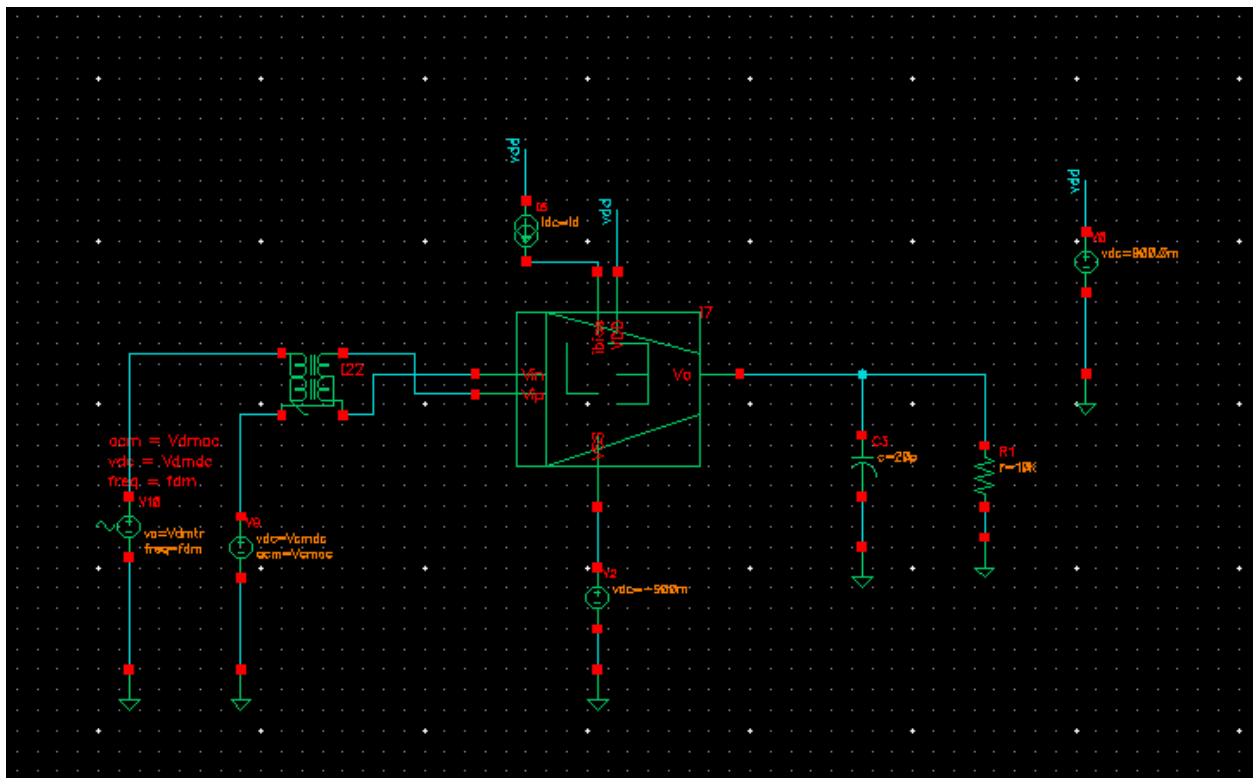
| Transistor number | Width | Length |
|-------------------|-------|--------|
| M1                | 9u    | 180n   |
| M2                | 9u    | 180n   |
| M3                | 20u   | 180n   |
| M4                | 20u   | 180n   |
| M5                | 10u   | 180n   |
| M6                | 10u   | 180n   |
| M7                | 10u   | 180n   |
| M8                | 185u  | 180n   |

Circuit Schematic Setup:

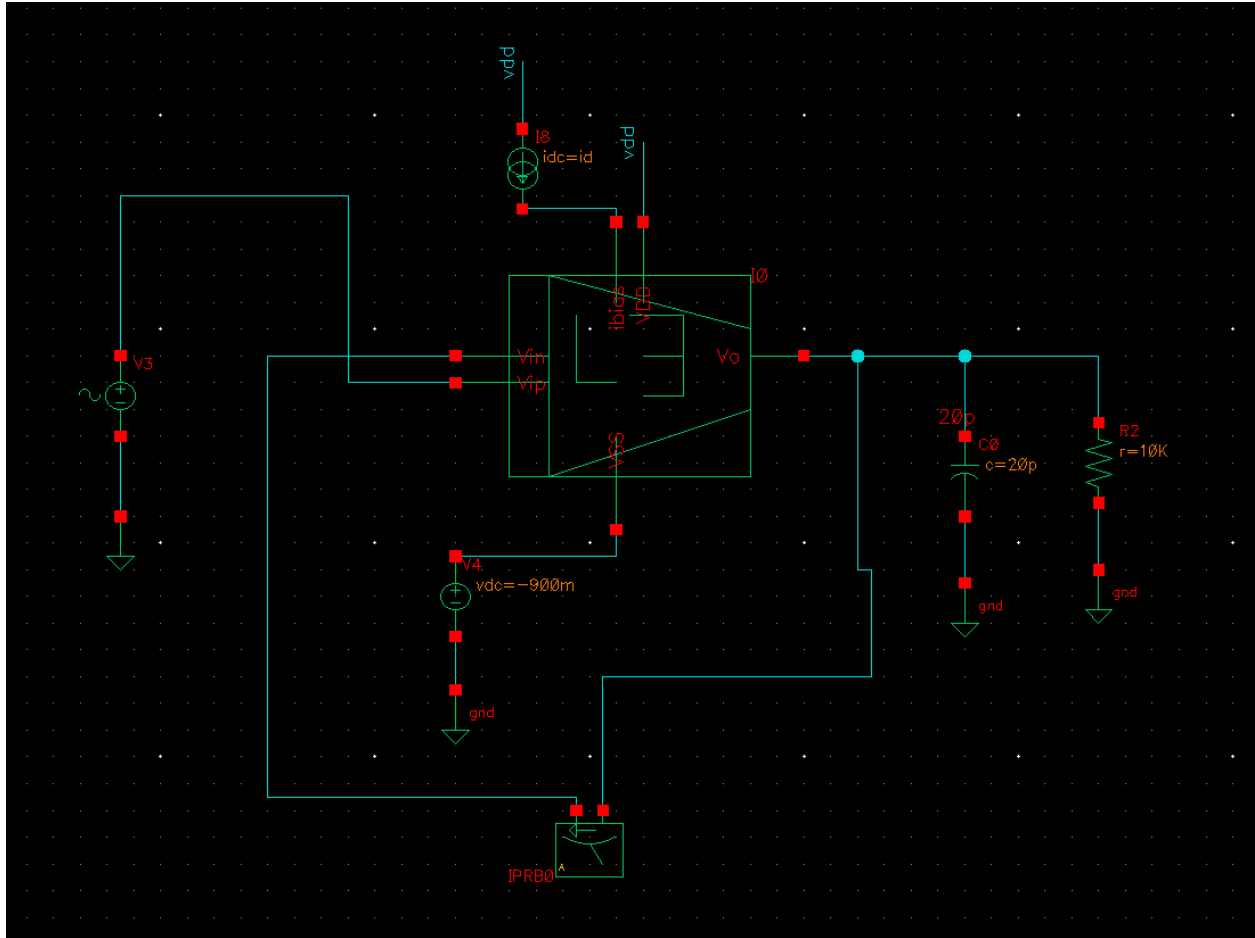


Circuit Test bench Set-up:

Open-loop:



Closed-loop setup:



ADL\_TEST SETUPS:

ADE XL Test Editor - Lab-3-607:lab-3\_tb:1@n01-zeus.olympus.ece.tamu.edu

Session Setup Analyses Variables Outputs Simulation Results Tools Help

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Design Variables

|   | Name    | Value |
|---|---------|-------|
| 1 | id      | 50u   |
| 2 | Vdmdc   | 0     |
| 3 | Vdmac   | 1     |
| 4 | Vdmtr   | 0     |
| 5 | fdm     | 0     |
| 6 | Vcmdc   | 0     |
| 7 | Vcmac   | 1     |
| 8 | p5vonly | 0     |

Analyses

| Type  | Enable                              | Arguments                                             |
|-------|-------------------------------------|-------------------------------------------------------|
| 1 stb | <input checked="" type="checkbox"/> | 1 100G 20 /IPRB0 Logarithmic Points Per Decade Sta... |

Outputs

| Name/Signal/Expr  | Value | Plot                                | Save                     | Save Options |
|-------------------|-------|-------------------------------------|--------------------------|--------------|
| 1 Loop Gain Phase | wave  | <input checked="" type="checkbox"/> | <input type="checkbox"/> |              |
| 2 Loop Gain dB20  | wave  | <input checked="" type="checkbox"/> | <input type="checkbox"/> |              |

> Results in ...ab-3-607:lab-3\_tb:1/simulation/lab

6(9) Debug Test | Status: Plotting results... | T=27 C | Simulator: spectre aps | State: Lab-3-607:lab-3\_tb:1\_active

ADL\_TEST\_SETUPS:

ADE XL Test Editor - dc@n01-zeus.olympus.ece.tamu.edu

Session Setup Analyses Variables Outputs Simulation Results Tools Help

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Design Variables

| Name      | Value |
|-----------|-------|
| 1 id      | 90u   |
| 2 Vdmdc   | 0     |
| 3 Vdmac   | 1     |
| 4 Vdmtr   | 0     |
| 5 fdm     | 0     |
| 6 Vcmdc   | 0     |
| 7 Vcmac   | 1     |
| 8 p5vonly | 0     |

Analyses

| Type | Enable                              | Arguments                                          |
|------|-------------------------------------|----------------------------------------------------|
| 1 dc | <input checked="" type="checkbox"/> | t -900m 900m 100u Linear Step Size Start-Stop /V9  |
| 2 ac | <input checked="" type="checkbox"/> | 1 100G 20 Logarithmic Points Per Decade Start-Stop |

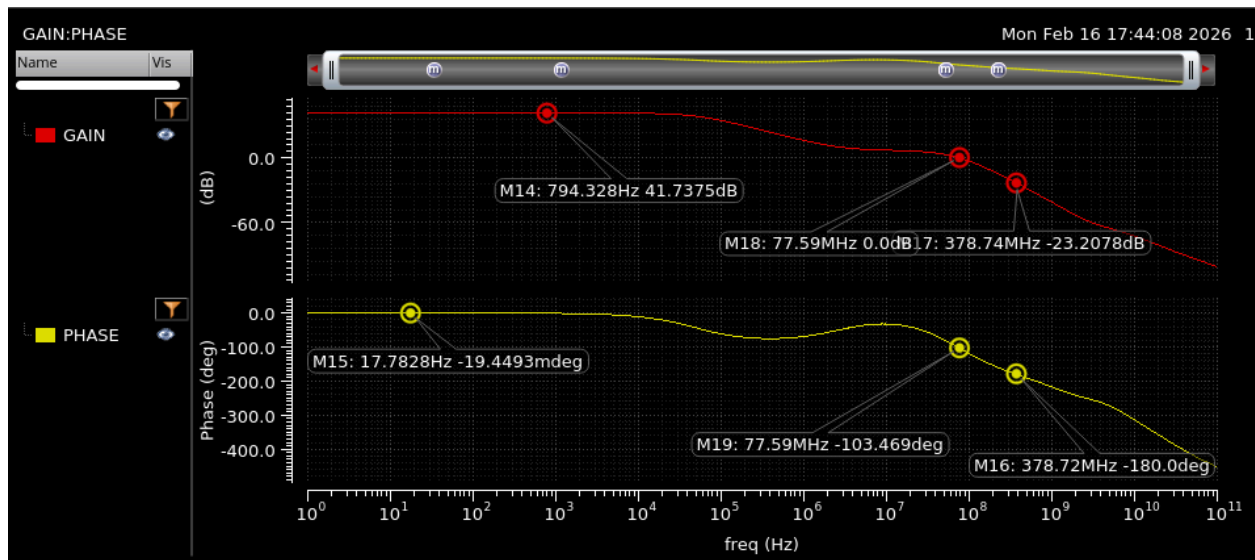
Outputs

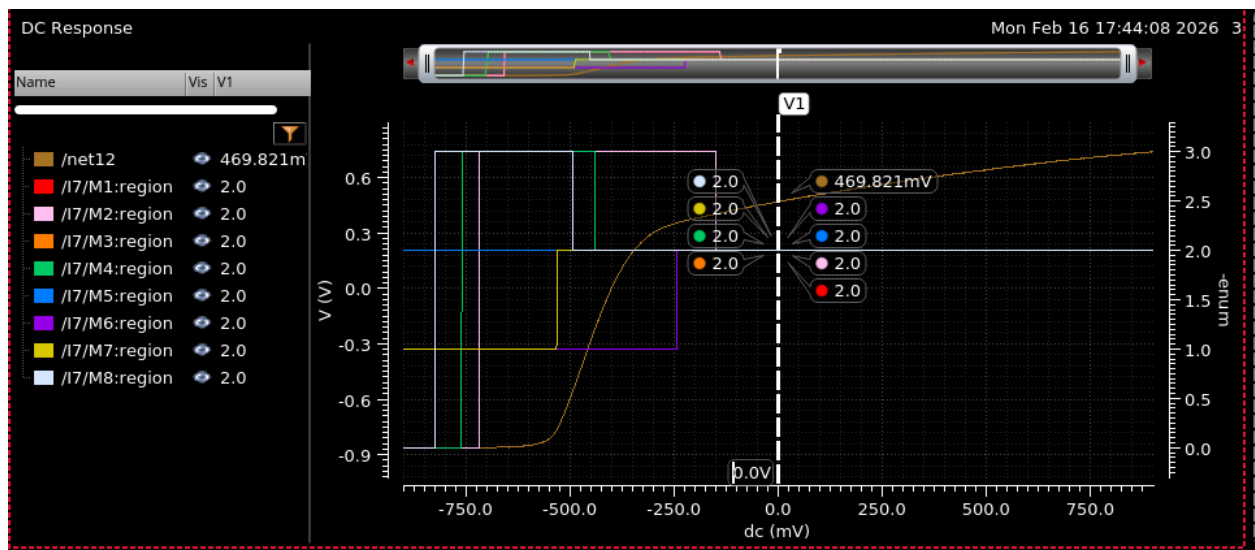
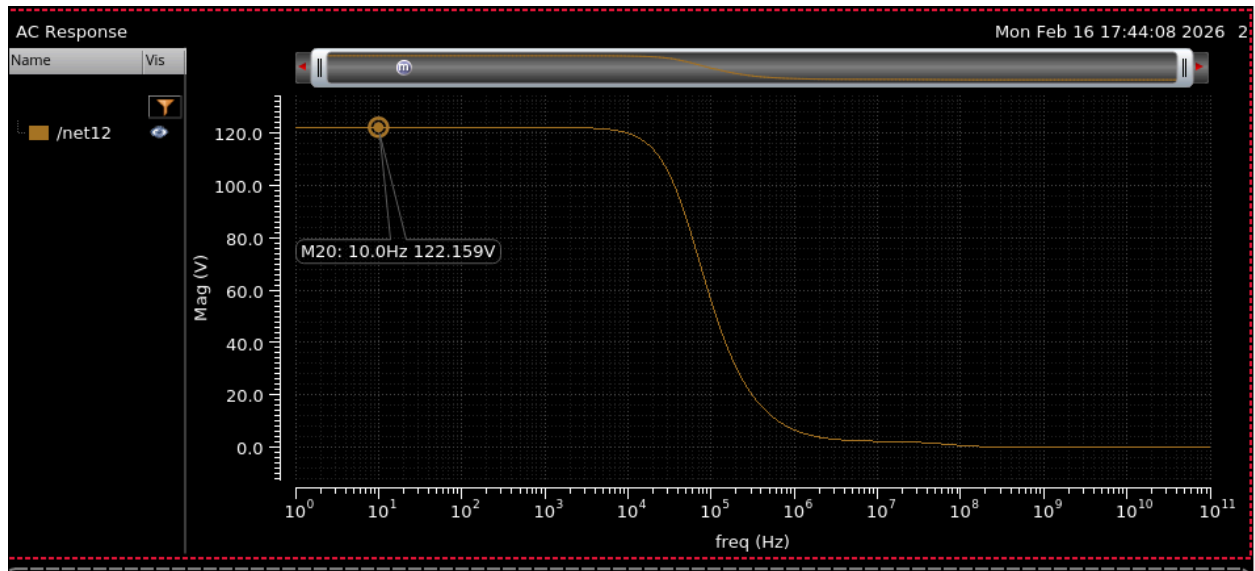
| Name/Signal/Expr | Value | Plot                                | Save                                | Save Options |
|------------------|-------|-------------------------------------|-------------------------------------|--------------|
| 1 GAIN           | wave  | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |              |
| 2 PHASE          | wave  | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |              |
| 3 net12          |       | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | allv         |

> Results in ...mpADEDir\_shes123/dc/simulation

4(5) | Temperature | Status: Ready | T=27 C | Simulator: spectre aps | State: dc\_active

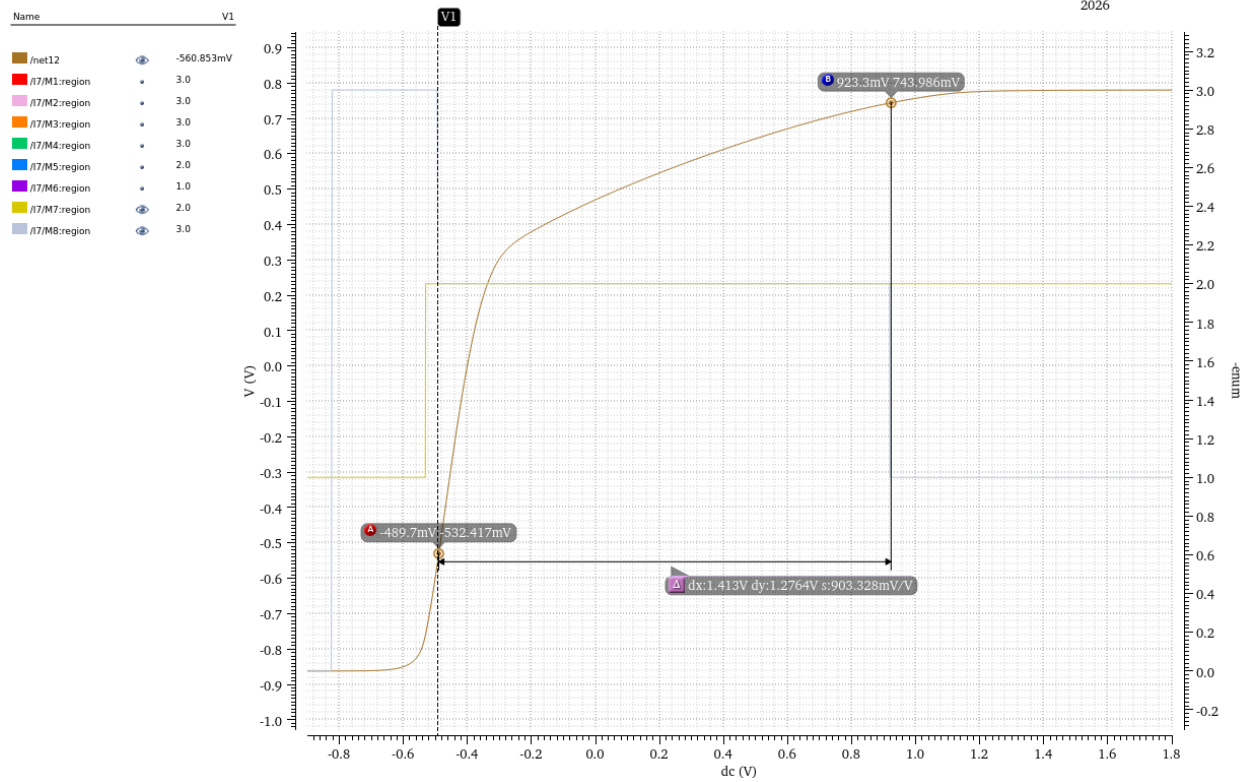
DC RUN-SATURATION,REGION ANALYSIS:(OPEN LOOP GAIN ANALYSIS)



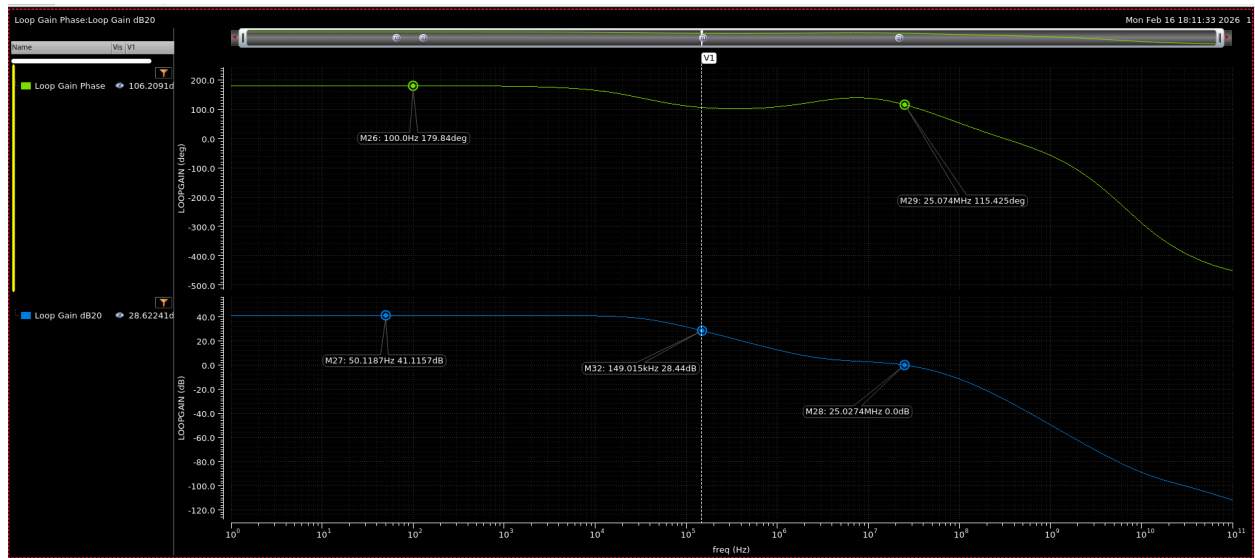


Output swing:





## CLOSED LOOP GAIN ANALYSIS:



## Gain Margin,Phase Margin Analysis:

ADE XL Test Editor - stb-closed-loop@n01-zeus.olympus.ece.tamu.edu

Session Setup Analyses Variables Outputs Simulation Results Tools Help

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Design Variables

| Name      | Value |
|-----------|-------|
| 1 id      | 90u   |
| 2 Vdmdc   | 0     |
| 3 Vdmac   | 1     |
| 4 Vdmtr   | 0     |
| 5 fdm     | 0     |
| 6 Vcmdc   | 0     |
| 7 Vcmac   | 1     |
| 8 p5vonly | 0     |

Analyses

| Type  | Enable                              | Arguments                                             |
|-------|-------------------------------------|-------------------------------------------------------|
| 1 stb | <input checked="" type="checkbox"/> | 1 100G 20 /IPRB0 Logarithmic Points Per Decade Sta... |

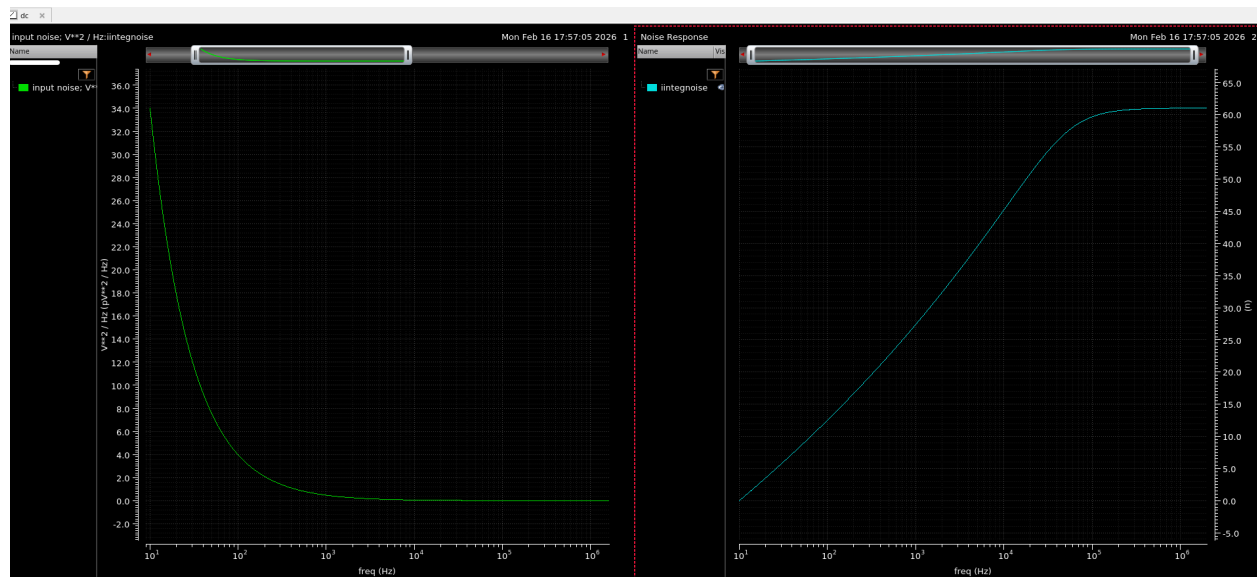
Outputs

| Name/Signal/Expr  | Value  | Plot                                | Save                                | Save Option |
|-------------------|--------|-------------------------------------|-------------------------------------|-------------|
| 1 Loop Gain Phase |        | <input type="checkbox"/>            | <input type="checkbox"/>            |             |
| 2 Loop Gain dB20  | wave   | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |             |
| 3 Gain Margin     | 24.23  | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> |             |
| 4 Phase Margin    | 83.76  | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> |             |
| 5 GAIN            | 41.81  | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> |             |
| 6 GBW             | 63.47M | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |             |

> Results in ...3\_tb/adexl/results/data/Interactive

12(25) Debug Test | Status: Ready | T=27 C | Simulator: spectre aps | State: stb-closed-loop\_active

## NOISE ANALYSIS:



ADE XL Test Editor - dc@n01-zeus.olympus.ece.tamu.edu

Session Setup Analyses Variables Outputs Simulation Results Tools Help

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Design Variables

|   | Name    | Value |
|---|---------|-------|
| 1 | id      | 90u   |
| 2 | Vdmdc   | 0     |
| 3 | Vdmac   | 1     |
| 4 | Vdmtr   | 0     |
| 5 | fdm     | 0     |
| 6 | Vcmdc   | 0     |
| 7 | Vcmac   | 1     |
| 8 | p5vonly | 0     |

Analyses

| Type    | Enable                              | Arguments                                             |
|---------|-------------------------------------|-------------------------------------------------------|
| 1 dc    | <input type="checkbox"/>            | t -900m 900m 100u Linear Step Size Start-Stop /V9     |
| 2 ac    | <input type="checkbox"/>            | 1 100G 20 Logarithmic Points Per Decade Start-Stop    |
| 3 noise | <input checked="" type="checkbox"/> | 10 2M 100 Logarithmic Points Per Decade Start-Stop... |

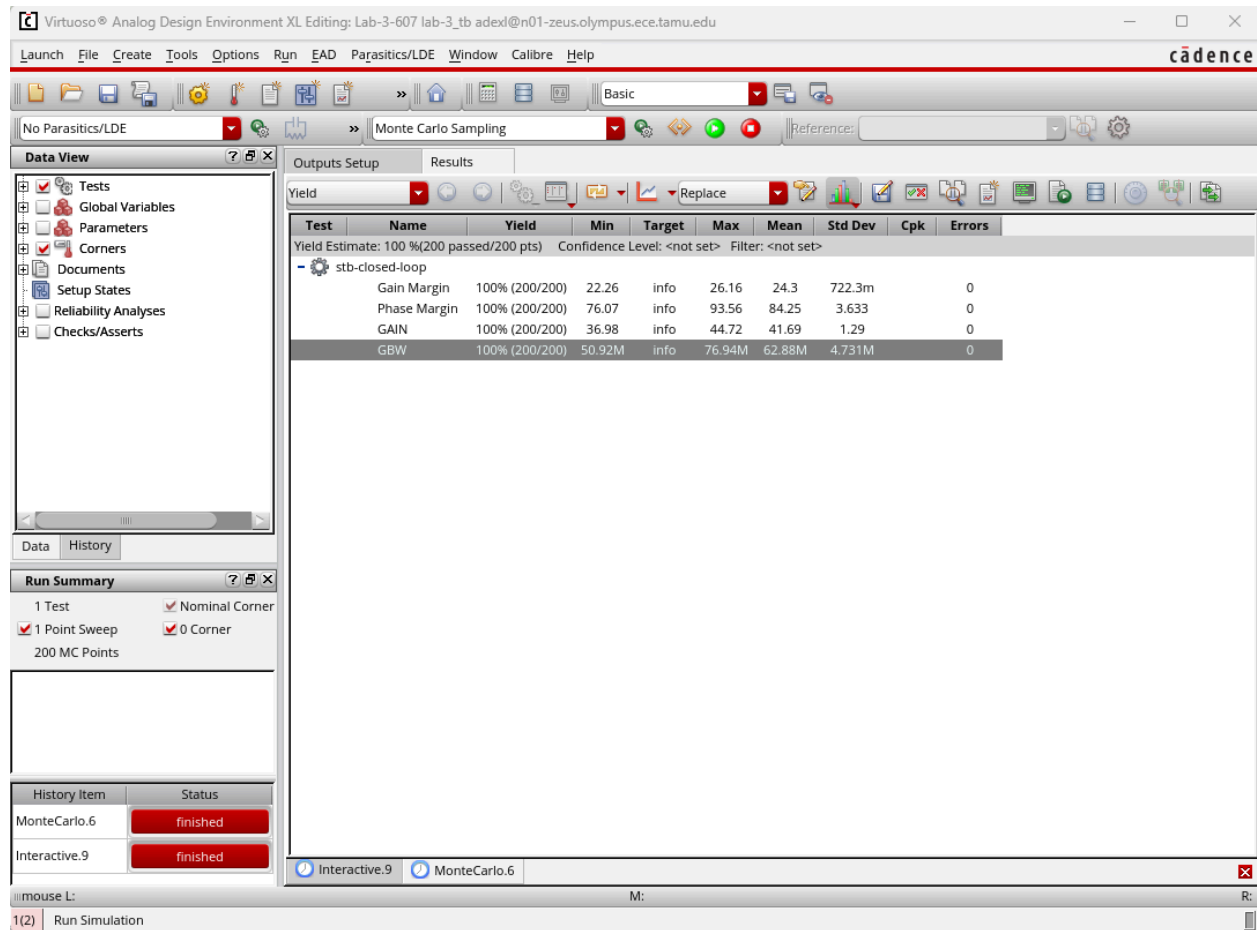
Outputs

| Name/Signal/Expr         | Value  | Plot                                | Save                                | Save Options |
|--------------------------|--------|-------------------------------------|-------------------------------------|--------------|
| 1 GAIN                   |        | <input type="checkbox"/>            | <input type="checkbox"/>            |              |
| 2 PHASE                  |        | <input type="checkbox"/>            | <input type="checkbox"/>            |              |
| 3 net12                  |        | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | allv         |
| 4 input noise; V**2 / Hz | wave   | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |              |
| 5 iintegnoise            | wave   | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |              |
| 6 INTEH_NOISE            | 61.04u | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |              |
| 7 SQRT(NOISE)            | 7.813m | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |              |

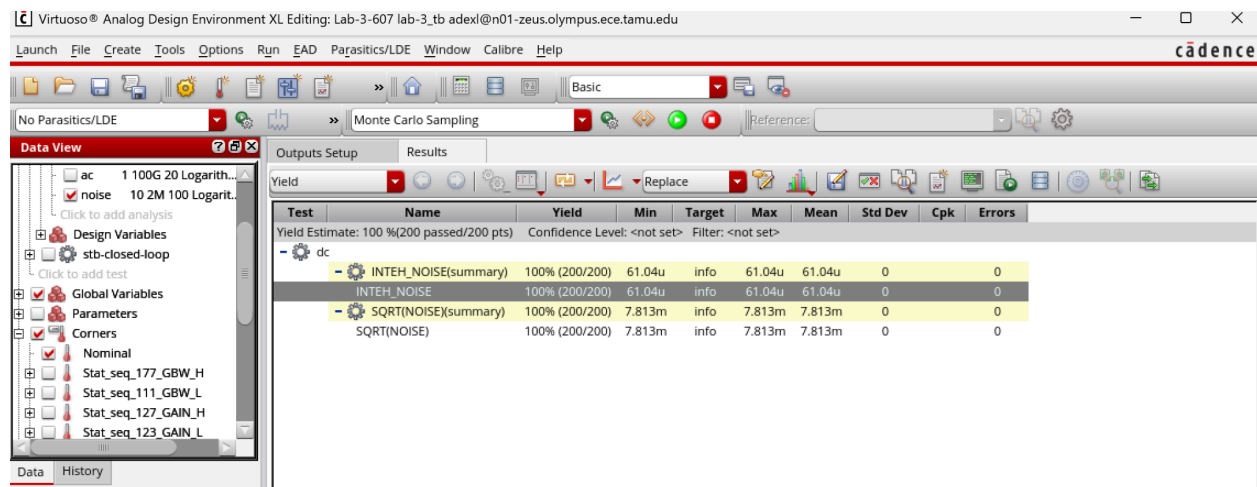
> Results in ...mpADEDir\_shes123/dc/simulation/lab-3\_tb/spectre/schematic

4(5) Simulator ... | Status: Ready | T=27 C | Simulator: spectre aps | State: dc\_active

2. Run brute-force MC simulations and update the best & worst corners that you created in Lab1 with information from best and worst 3-sigma noise values.



### Noise values:



3. Report the DC Gain, GBW, Phase Margin and Gain Margin for all corners. (No plots needed, just report the numbers in a comprehensive table).

| Test            | Output       | Nominal | Spec | Weight | Pass/Fail | Min    | Max    | Stat_seq_177_GBW_H | Stat_seq_111_GBW_L |
|-----------------|--------------|---------|------|--------|-----------|--------|--------|--------------------|--------------------|
| stb-closed-loop | Gain Margin  | 24.23   |      |        |           | 22.26  | 26.16  | 23.33              | 25.43              |
| stb-closed-loop | Phase Margin | 83.76   |      |        |           | 76.07  | 93.56  | 77.81              | 93.56              |
| stb-closed-loop | GAIN         | 41.81   |      |        |           | 36.98  | 44.72  | 38.58              | 39.81              |
| stb-closed-loop | GBW          | 63.47M  |      |        |           | 50.92M | 76.94M | 76.94M             | 50.92M             |

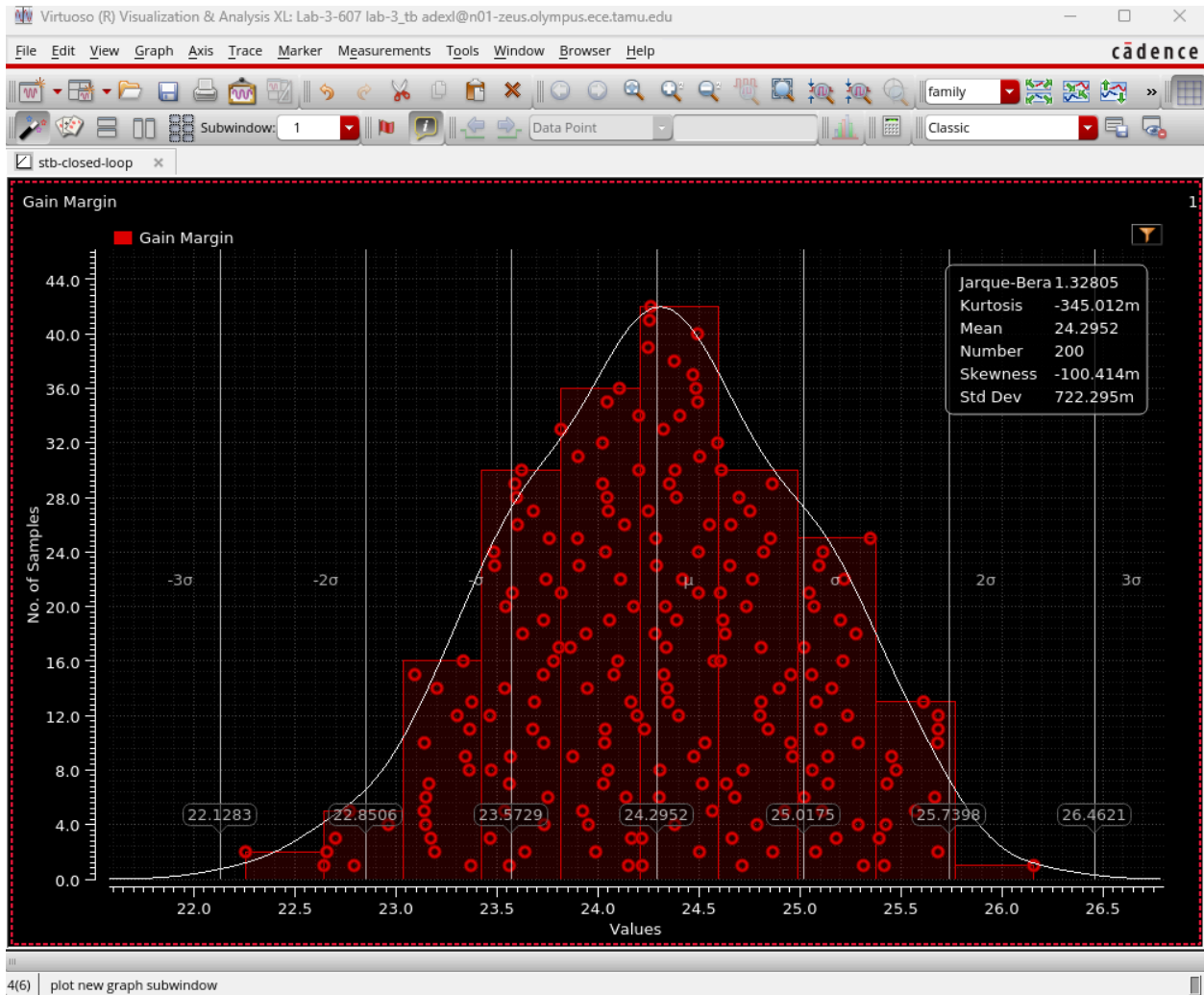
| Stat_seq_127_GAIN_H | Stat_seq_123_GAIN_L | Stat_seq_111_Phase_Margin_H | Stat_seq_96_Phase_Margin_L |
|---------------------|---------------------|-----------------------------|----------------------------|
| 24.39               | 23.73               | 25.43                       | 22.26                      |
| 83.08               | 84.63               | 93.56                       | 76.07                      |
| 44.72               | 36.98               | 39.81                       | 42.07                      |
| 61.93M              | 64.27M              | 50.92M                      | 66.06M                     |

| Stat_seq_96_Phase_Margin_L | Stat_seq_76_Gain_Margin_H | Stat_seq_96_Gain_Margin_L |
|----------------------------|---------------------------|---------------------------|
| 22.26                      | 26.16                     | 22.26                     |
| 76.07                      | 93.06                     | 76.07                     |
| 42.07                      | 42.68                     | 42.07                     |
| 66.06M                     | 52.33M                    | 66.06M                    |

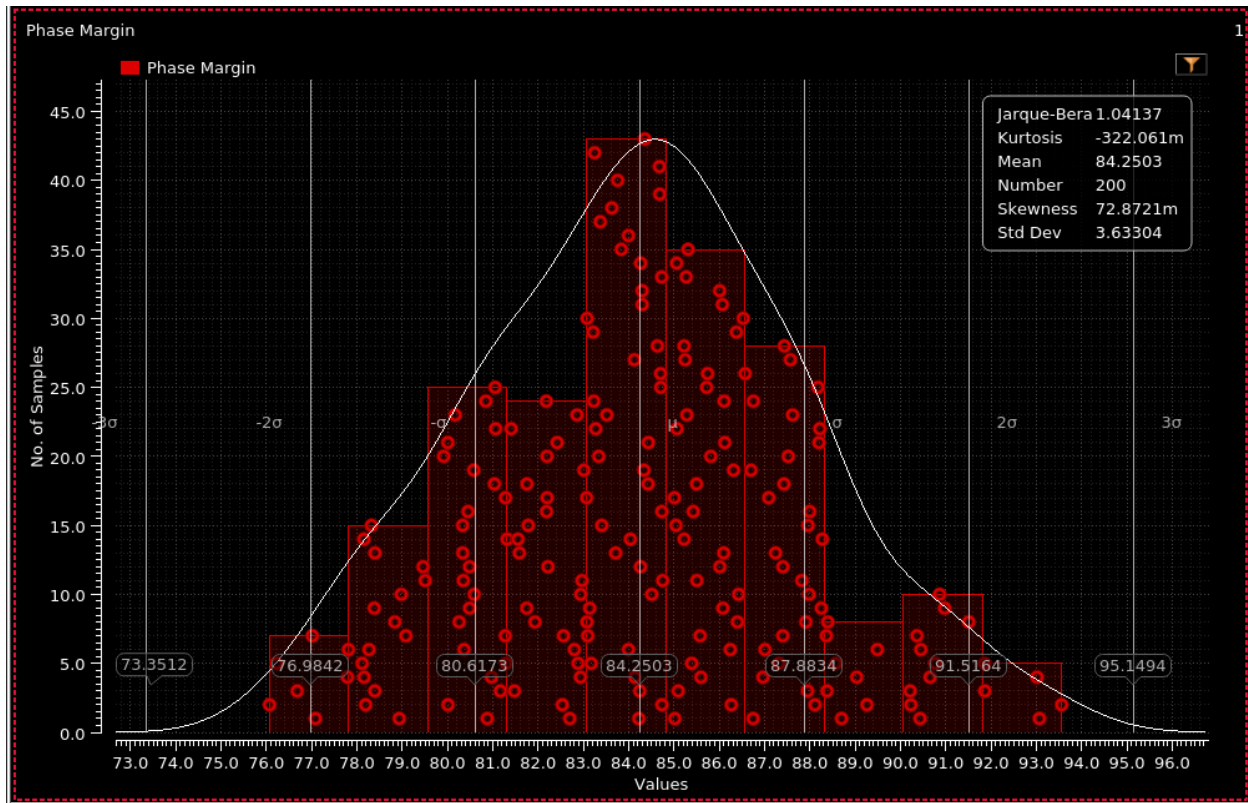
4. Adjust your design to target a 3-sigma yield for DC Gain, GBW, Phase Margin (Show histograms).

Meeting the specs in the first run:

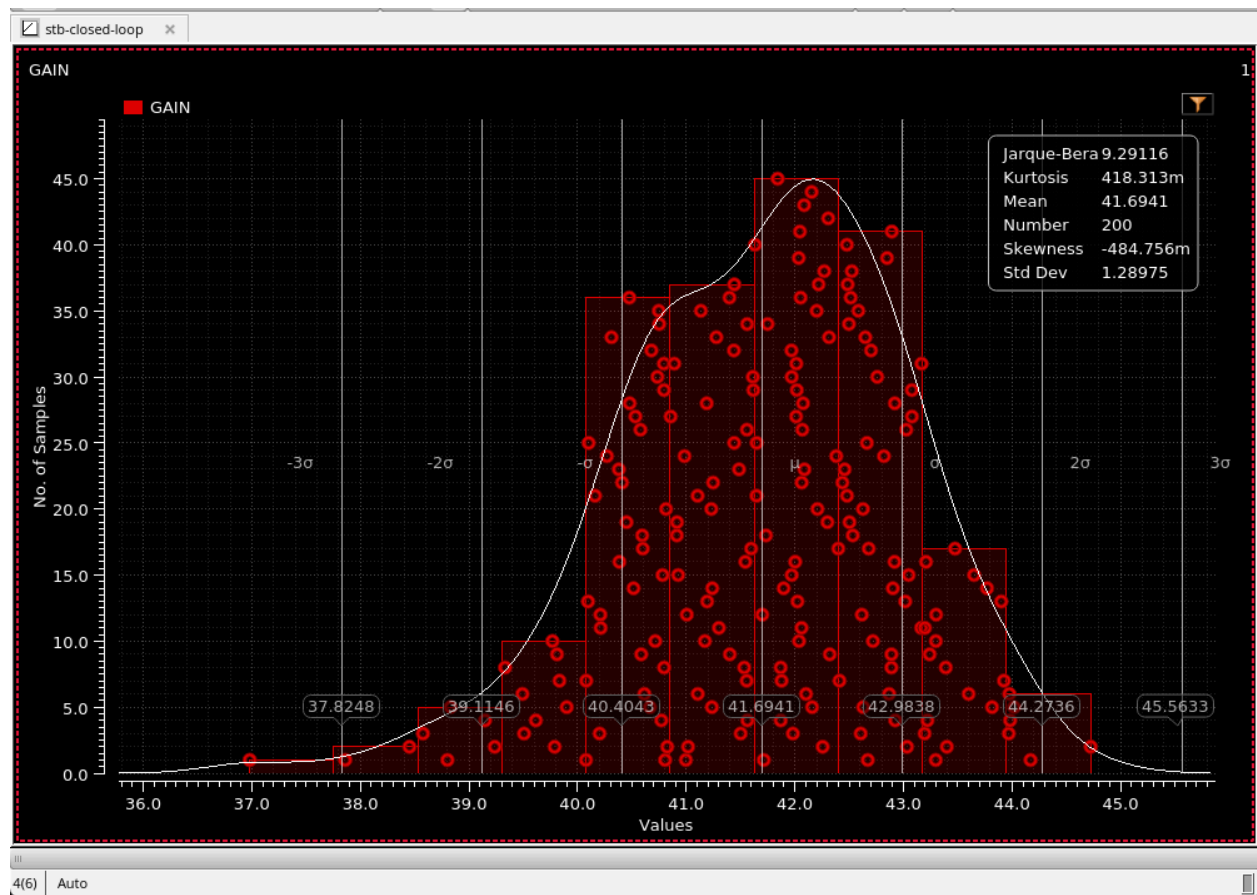
Gain Margin:



Phase Margin:

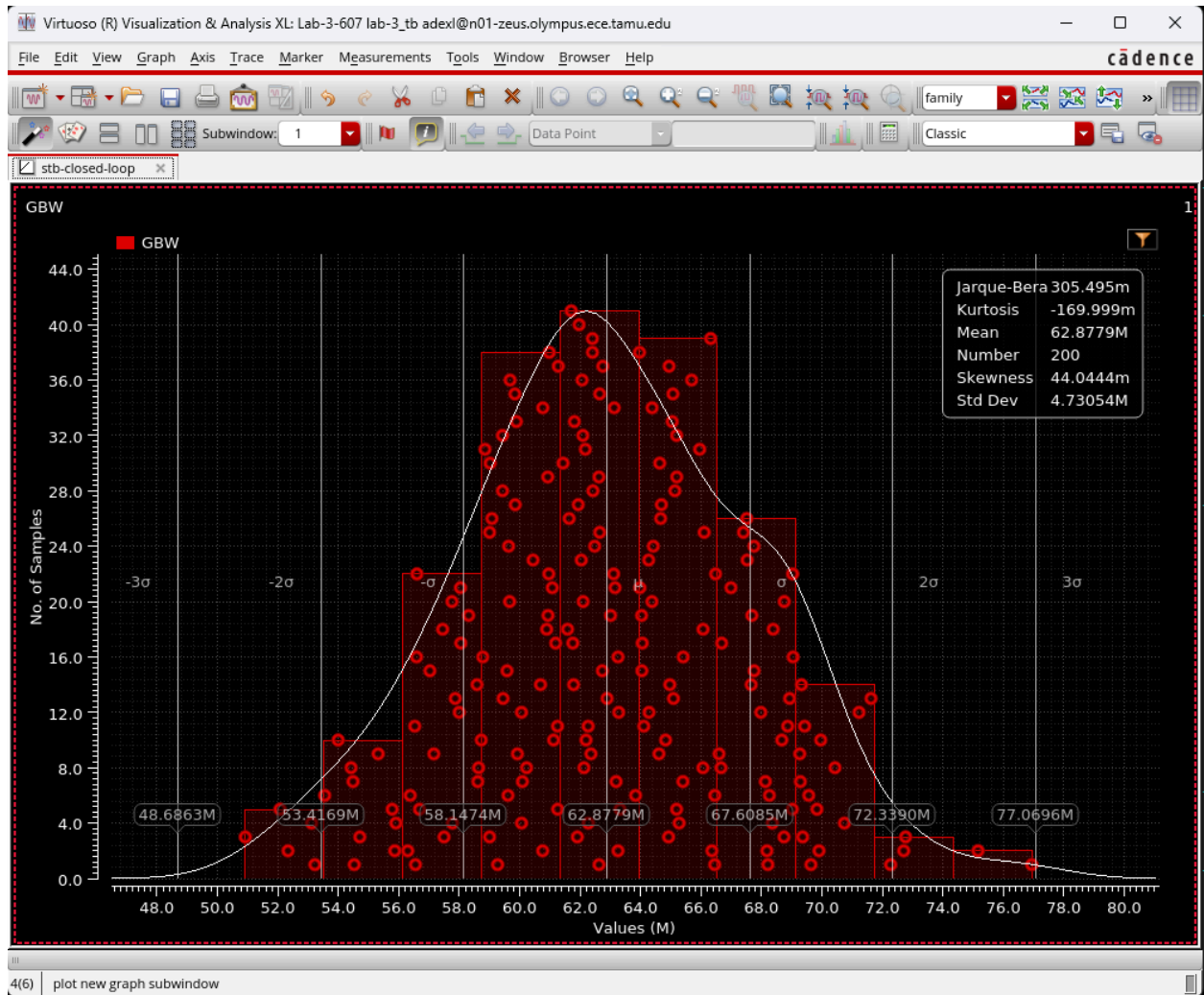


Gain:

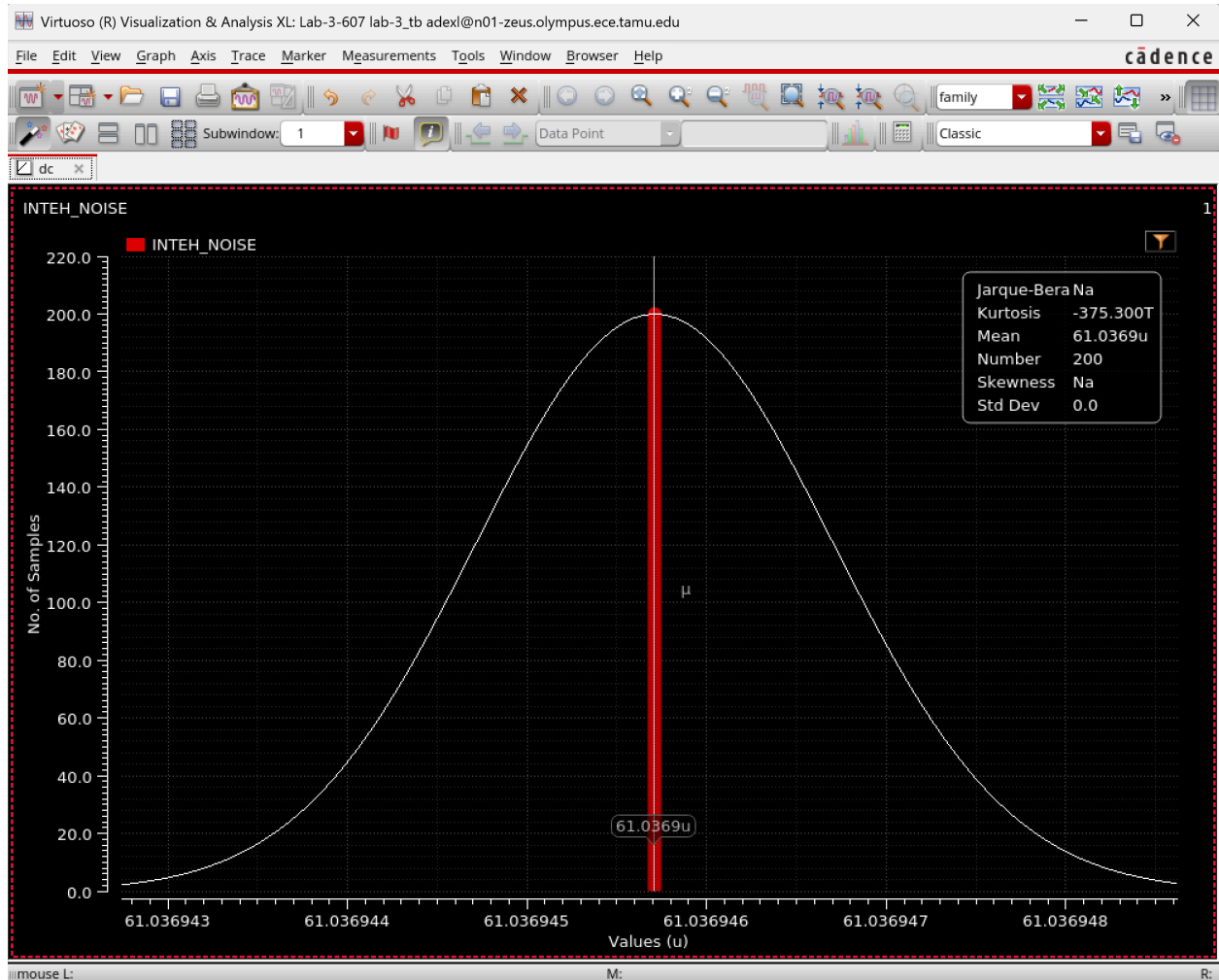


GBW:





For noise histogram:



5. Do a cross simulation (two corners, supply variation and temperature variation) and report the worst case

DC Gain, GBW, Phase Margin and input referred noise. Assume there is +/- 5% variation on the supply,

-40 < temperature < 125 C. You do not have to adjust the design if the specs are not met.

Provide comparison tables of hand-calculated vs. final transistor sizes, and required specs vs. simulated

Specs.-----

Selecting two corners as the Gain high and Gain low:

Displaying the graphs that are obtained by sweeping the temperature and the vdd value



**Single Run, Sweeps and Corners**

**Data View**

- ☒ fdm
- ☒ Vcmdd
- ☒ Vcmac
- ☒ VDD
- ☒ id
- ☒ temperature

**Parameters**

- ☒ Nominal
- ☒ Stat\_seq\_177\_GBW\_H
- ☒ Stat\_seq\_111\_GBW\_L
- ☒ Stat\_seq\_127\_GAIN\_H
- ☒ Stat\_seq\_123\_GAIN\_L
- ☒ Stat\_seq\_111\_Phase\_Margin
- ☒ Stat\_seq\_96\_Phase\_Margin
- ☒ Stat\_seq\_76\_Gain\_Margin\_H
- ☒ Stat\_seq\_96\_Gain\_Margin\_L

**Documents**

- ☒ Setup States
- ☒ Reliability Analyses
- ☒ Checks/Asserts

**Run Summary**

1 Test ☒ Nominal Corner

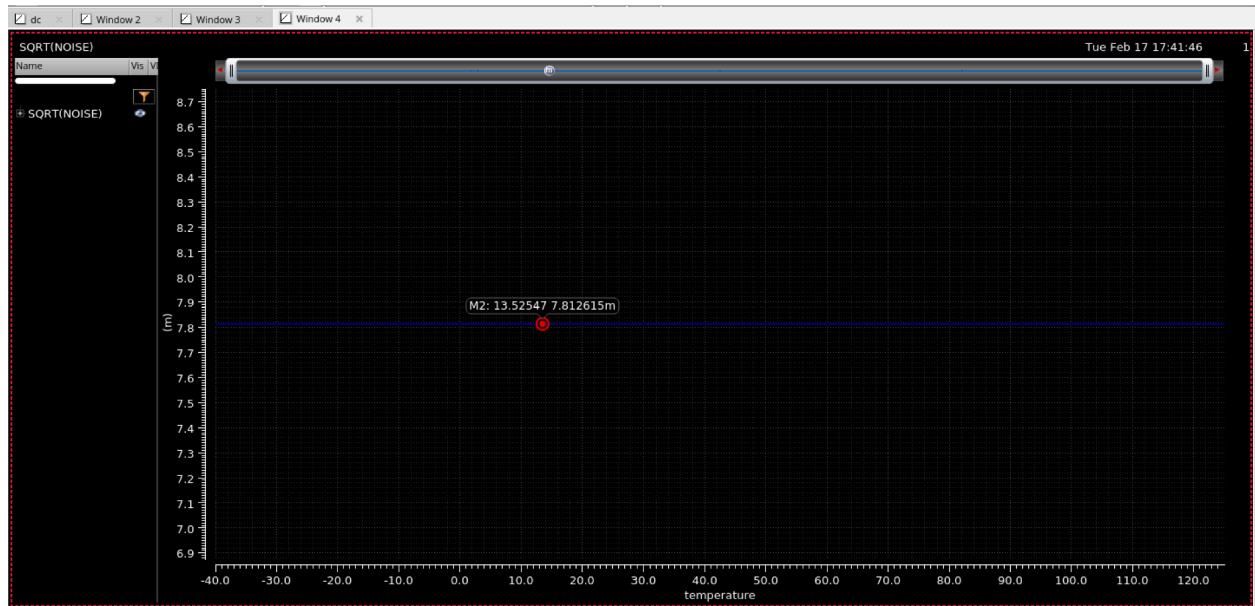
☒ 30 Point Sweeps ☒ 2 Corners

| Point                                   | Test            | Output       | Nominal | Spec | Weight | Pass/Fail | Min    | Max    | Stat_seq_127_GAIN_H | Stat_seq_123_GAIN_L |
|-----------------------------------------|-----------------|--------------|---------|------|--------|-----------|--------|--------|---------------------|---------------------|
| Parameters: VDD=900m, temperature=0     |                 |              |         |      |        |           |        |        |                     |                     |
| 1                                       | stb-closed-loop | Gain Margin  | 23.5    |      |        |           | 23.07  | 23.62  | 23.62               | 23.07               |
| 1                                       | stb-closed-loop | Phase Margin | 79.08   |      |        |           | 78.24  | 80.22  | 78.24               | 80.22               |
| 1                                       | stb-closed-loop | GAIN         | 42.28   |      |        |           | 37.35  | 45.28  | 45.28               | 37.35               |
| 1                                       | stb-closed-loop | GBW          | 71.06M  |      |        |           | 69.64M | 71.38M | 69.64M              | 71.38M              |
| Parameters: VDD=900m, temperature=-40   |                 |              |         |      |        |           |        |        |                     |                     |
| 2                                       | stb-closed-loop | Gain Margin  | 22.35   |      |        |           | 22.04  | 22.39  | 22.39               | 22.04               |
| 2                                       | stb-closed-loop | Phase Margin | 71.94   |      |        |           | 70.84  | 73.6   | 70.84               | 73.6                |
| 2                                       | stb-closed-loop | GAIN         | 42.97   |      |        |           | 37.89  | 46.1   | 46.1                | 37.89               |
| 2                                       | stb-closed-loop | GBW          | 83.92M  |      |        |           | 82.73M | 83.92M | 82.73M              | 83.24M              |
| Parameters: VDD=900m, temperature=1.25  |                 |              |         |      |        |           |        |        |                     |                     |
| 3                                       | stb-closed-loop | Gain Margin  | 23.54   |      |        |           | 23.1   | 23.65  | 23.65               | 23.1                |
| 3                                       | stb-closed-loop | Phase Margin | 79.3    |      |        |           | 78.47  | 80.43  | 78.47               | 80.43               |
| 3                                       | stb-closed-loop | GAIN         | 42.25   |      |        |           | 37.33  | 45.26  | 45.26               | 37.33               |
| 3                                       | stb-closed-loop | GBW          | 70.69M  |      |        |           | 69.26M | 71.03M | 69.26M              | 71.03M              |
| Parameters: VDD=900m, temperature=42.5  |                 |              |         |      |        |           |        |        |                     |                     |
| 4                                       | stb-closed-loop | Gain Margin  | 24.63   |      |        |           | 24.1   | 24.81  | 24.81               | 24.1                |
| 4                                       | stb-closed-loop | Phase Margin | 86.39   |      |        |           | 85.79  | 87.12  | 85.79               | 87.12               |
| 4                                       | stb-closed-loop | GAIN         | 41.53   |      |        |           | 36.77  | 44.4   | 44.4                | 36.77               |
| 4                                       | stb-closed-loop | GBW          | 59.48M  |      |        |           | 57.89M | 60.48M | 57.89M              | 60.48M              |
| Parameters: VDD=900m, temperature=83.75 |                 |              |         |      |        |           |        |        |                     |                     |
| 5                                       | stb-closed-loop | Gain Margin  | 25.66   |      |        |           | 25.02  | 25.89  | 25.89               | 25.02               |
| 5                                       | stb-closed-loop | Phase Margin | 93.13   |      |        |           | 92.71  | 93.57  | 92.71               | 93.57               |
| 5                                       | stb-closed-loop | GAIN         | 40.78   |      |        |           | 36.19  | 43.5   | 43.5                | 36.19               |
| 5                                       | stb-closed-loop | GBW          | 49.92M  |      |        |           | 48.26M | 51.38M | 48.26M              | 51.38M              |
| Parameters: VDD=900m, temperature=125   |                 |              |         |      |        |           |        |        |                     |                     |
| 6                                       | stb-closed-loop | Gain Margin  | 26.62   |      |        |           | 25.9   | 26.91  | 26.91               | 25.9                |
| 6                                       | stb-closed-loop | Phase Margin | 99.49   |      |        |           | 99.23  | 99.73  | 99.23               | 99.73               |
| 6                                       | stb-closed-loop | GAIN         | 39.99   |      |        |           | 35.58  | 42.53  | 42.53               | 35.58               |
| 6                                       | stb-closed-loop | GBW          | 41.79M  |      |        |           | 40.08M | 43.53M | 40.08M              | 43.53M              |
| Parameters: VDD=810m, temperature=0     |                 |              |         |      |        |           |        |        |                     |                     |
| 7                                       | stb-closed-loop | Gain Margin  | 23.88   |      |        |           | 23.59  | 23.93  | 23.93               | 23.59               |
| 7                                       | stb-closed-loop | Phase Margin | 82.33   |      |        |           | 80.83  | 85.1   | 80.83               | 85.1                |
| 7                                       | stb-closed-loop | GAIN         | 42.15   |      |        |           | 37.3   | 45.11  | 45.11               | 37.3                |
| 7                                       | stb-closed-loop | GBW          | 65.46M  |      |        |           | 63.69M | 65.46M | 65.06M              | 63.69M              |
| Parameters: VDD=810m, temperature=-40   |                 |              |         |      |        |           |        |        |                     |                     |
| 8                                       | stb-closed-loop | Gain Margin  | 22.77   |      |        |           | 22.61  | 22.77  | 22.74               | 22.61               |
| 8                                       | stb-closed-loop | Phase Margin | 75.25   |      |        |           | 73.48  | 78.58  | 73.48               | 78.58               |
| 8                                       | stb-closed-loop | GAIN         |         |      |        |           |        |        |                     |                     |

| Point                                   | Test            | Output       | Nominal | Spec | Weight | Pass/Fail | Min    | Max    | Stat_seq_127_GAIN_H | Stat_seq_123_GAIN_L |
|-----------------------------------------|-----------------|--------------|---------|------|--------|-----------|--------|--------|---------------------|---------------------|
| 8                                       | stb-closed-loop | GBW          | 77.35M  |      |        |           | 74.36M | 77.35M | 77.35M              | 74.36M              |
| Parameters: VDD=810m, temperature=1.25  |                 |              |         |      |        |           |        |        |                     |                     |
| 9                                       | stb-closed-loop | Gain Margin  | 23.92   |      |        |           | 23.62  | 23.96  | 23.96               | 23.62               |
| 9                                       | stb-closed-loop | Phase Margin | 82.55   |      |        |           | 81.06  | 85.31  | 81.06               | 85.31               |
| 9                                       | stb-closed-loop | GAIN         | 42.13   |      |        |           | 37.28  | 45.09  | 45.09               | 37.28               |
| 9                                       | stb-closed-loop | GBW          | 65.12M  |      |        |           | 63.38M | 65.12M | 64.71M              | 63.38M              |
| Parameters: VDD=810m, temperature=42.5  |                 |              |         |      |        |           |        |        |                     |                     |
| 10                                      | stb-closed-loop | Gain Margin  | 24.98   |      |        |           | 24.57  | 25.1   | 25.1                | 24.57               |
| 10                                      | stb-closed-loop | Phase Margin | 89.57   |      |        |           | 88.31  | 91.89  | 88.31               | 91.89               |
| 10                                      | stb-closed-loop | GAIN         | 41.39   |      |        |           | 36.7   | 44.21  | 44.21               | 36.7                |
| 10                                      | stb-closed-loop | GBW          | 54.68M  |      |        |           | 53.84M | 54.68M | 54M                 | 53.84M              |
| Parameters: VDD=810m, temperature=83.75 |                 |              |         |      |        |           |        |        |                     |                     |
| 11                                      | stb-closed-loop | Gain Margin  | 25.97   |      |        |           | 25.46  | 26.16  | 26.16               | 25.46               |
| 11                                      | stb-closed-loop | Phase Margin | 96.21   |      |        |           | 95.16  | 98.19  | 95.16               | 98.19               |
| 11                                      | stb-closed-loop | GAIN         | 40.63   |      |        |           | 36.1   | 43.31  | 43.31               | 36.1                |
| 11                                      | stb-closed-loop | GBW          | 45.8M   |      |        |           | 44.93M | 45.8M  | 44.93M              | 45.6M               |
| Parameters: VDD=810m, temperature=125   |                 |              |         |      |        |           |        |        |                     |                     |
| 12                                      | stb-closed-loop | Gain Margin  | 26.91   |      |        |           | 26.3   | 27.15  | 27.15               | 26.3                |
| 12                                      | stb-closed-loop | Phase Margin | 102.5   |      |        |           | 101.6  | 104.2  | 101.6               | 104.2               |
| 12                                      | stb-closed-loop | GAIN         | 39.82   |      |        |           | 35.48  | 42.32  | 42.32               | 35.48               |
| 12                                      | stb-closed-loop | GBW          | 38.24M  |      |        |           | 37.24M | 38.5M  | 37.24M              | 38.5M               |
| Parameters: VDD=855m, temperature=0     |                 |              |         |      |        |           |        |        |                     |                     |
| 13                                      | stb-closed-loop | Gain Margin  | 23.69   |      |        |           | 23.32  | 23.77  | 23.77               | 23.32               |
| 13                                      | stb-closed-loop | Phase Margin | 80.65   |      |        |           | 79.49  | 82.56  | 79.49               | 82.56               |
| 13                                      | stb-closed-loop | GAIN         | 42.22   |      |        |           | 37.33  | 45.2   | 45.2                | 37.33               |
| 13                                      | stb-closed-loop | GBW          | 68.3M   |      |        |           | 67.39M | 68.3M  | 67.39M              | 67.57M              |
| Parameters: VDD=855m, temperature=-40   |                 |              |         |      |        |           |        |        |                     |                     |
| 14                                      | stb-closed-loop | Gain Margin  | 22.55   |      |        |           | 22.32  | 22.56  | 22.56               | 22.32               |
| 14                                      | stb-closed-loop | Phase Margin | 73.54   |      |        |           | 72.11  | 75.98  | 72.11               | 75.98               |
| 14                                      | stb-closed-loop | GAIN         | 42.92   |      |        |           | 37.87  | 46.03  | 46.03               | 37.87               |
| 14                                      | stb-closed-loop | GBW          | 80.66M  |      |        |           | 78.83M | 80.66M | 80.06M              | 78.83M              |
| Parameters: VDD=855m, temperature=1.25  |                 |              |         |      |        |           |        |        |                     |                     |
| 15                                      | stb-closed-loop | Gain Margin  | 23.72   |      |        |           | 23.35  | 23.8   | 23.8                | 23.35               |
| 15                                      | stb-closed-loop | Phase Margin | 80.87   |      |        |           | 79.72  | 82.77  | 79.72               | 82.77               |
| 15                                      | stb-closed-loop | GAIN         | 42.2    |      |        |           | 37.31  | 45.18  | 45.18               | 37.31               |
| 15                                      | stb-closed-loop | GBW          | 67.94M  |      |        |           | 67.03M | 67.94M | 67.03M              | 67.24M              |
| Parameters: VDD=855m, temperature=42.5  |                 |              |         |      |        |           |        |        |                     |                     |
| 16                                      | stb-closed-loop | Gain Margin  | 24.8    |      |        |           | 24.33  | 24.95  | 24.95               | 24.33               |
| 16                                      | stb-closed-loop | Phase Margin | 87.93   |      |        |           | 87.01  | 89.42  | 87.01               | 89.42               |
| 16                                      | stb-closed-loop | GAIN         | 41.47   |      |        |           | 36.74  | 44.31  | 44.31               | 36.74               |
| 16                                      | stb-closed-loop | GBW          | 57.1M   |      |        |           | 55.96M | 57.19M | 55.96M              | 57.19M              |
| Parameters: VDD=855m, temperature=83.75 |                 |              |         |      |        |           |        |        |                     |                     |
| 17                                      | stb-closed-loop | Gain Margin  | 25.81   |      |        |           | 25.24  | 26.02  | 26.02               | 25.24               |

Running corners for noise:





Comparison table showing the calculated and obtained specifications:

| Calculated width | Transistor number | Vs Simulated width |
|------------------|-------------------|--------------------|
| 1u               | M1                | 9u                 |
| 1u               | M2                | 9u                 |
| 4u               | M3                | 20u                |
| 4u               | M4                | 20u                |
| 2u               | M5                | 10u                |
| 2u               | M6                | 10u                |
| 2u               | M7                | 10u                |
| 4u               | M8                | 185u               |

| Point                                   | Test            | Output       | Nominal | Spec | Weight | Pass/Fail | Min    | Max    | Stat_seq_127_GAIN_H | Stat_seq_123_GAIN_L |
|-----------------------------------------|-----------------|--------------|---------|------|--------|-----------|--------|--------|---------------------|---------------------|
| 22                                      | stb-closed-loop | Phase Margin | 84.95   |      |        |           | 84.64  | 84.99  | 84.64               | 84.99               |
| 22                                      | stb-closed-loop | GAIN         | 41.59   |      |        |           | 36.79  | 44.47  | 44.47               | 36.79               |
| 22                                      | stb-closed-loop | GBW          | 61.8M   |      |        |           | 59.76M | 63.72M | 59.76M              | 63.72M              |
| Parameters: VDD=945m, temperature=83.75 |                 |              |         |      |        |           |        |        |                     |                     |
| 23                                      | stb-closed-loop | Gain Margin  | 25.51   |      |        |           | 24.82  | 25.77  | 25.77               | 24.82               |
| 23                                      | stb-closed-loop | Phase Margin | 91.73   |      |        |           | 91.5   | 91.73  | 91.6                | 91.5                |
| 23                                      | stb-closed-loop | GAIN         | 40.84   |      |        |           | 36.21  | 43.58  | 43.58               | 36.21               |
| 23                                      | stb-closed-loop | GBW          | 51.93M  |      |        |           | 49.85M | 54.2M  | 49.85M              | 54.2M               |
| Parameters: VDD=945m, temperature=125   |                 |              |         |      |        |           |        |        |                     |                     |
| 24                                      | stb-closed-loop | Gain Margin  | 26.48   |      |        |           | 25.71  | 26.8   | 26.8                | 25.71               |
| 24                                      | stb-closed-loop | Phase Margin | 98.14   |      |        |           | 97.71  | 98.15  | 98.15               | 97.71               |
| 24                                      | stb-closed-loop | GAIN         | 40.06   |      |        |           | 35.61  | 42.61  | 42.61               | 35.61               |
| 24                                      | stb-closed-loop | GBW          | 43.51M  |      |        |           | 41.46M | 45.99M | 41.46M              | 45.99M              |
| Parameters: VDD=990m, temperature=0     |                 |              |         |      |        |           |        |        |                     |                     |
| 25                                      | stb-closed-loop | Gain Margin  | 23.15   |      |        |           | 22.6   | 23.33  | 23.33               | 22.6                |
| 25                                      | stb-closed-loop | Phase Margin | 76.21   |      |        |           | 75.96  | 76.21  | 75.96               | 76.03               |
| 25                                      | stb-closed-loop | GAIN         | 42.36   |      |        |           | 37.37  | 45.42  | 45.42               | 37.37               |
| 25                                      | stb-closed-loop | GBW          | 76.5M   |      |        |           | 74.02M | 78.84M | 74.02M              | 78.84M              |
| Parameters: VDD=990m, temperature=-40   |                 |              |         |      |        |           |        |        |                     |                     |
| 26                                      | stb-closed-loop | Gain Margin  | 21.97   |      |        |           | 21.53  | 22.08  | 22.08               | 21.53               |
| 26                                      | stb-closed-loop | Phase Margin | 69.04   |      |        |           | 68.54  | 69.39  | 68.54               | 69.39               |
| 26                                      | stb-closed-loop | GAIN         | 43.04   |      |        |           | 37.89  | 46.22  | 46.22               | 37.89               |
| 26                                      | stb-closed-loop | GBW          | 90.27M  |      |        |           | 87.83M | 91.87M | 87.83M              | 91.87M              |
| Parameters: VDD=990m, temperature=1.25  |                 |              |         |      |        |           |        |        |                     |                     |
| 27                                      | stb-closed-loop | Gain Margin  | 23.19   |      |        |           | 22.64  | 23.37  | 23.37               | 22.64               |
| 27                                      | stb-closed-loop | Phase Margin | 76.44   |      |        |           | 76.19  | 76.44  | 76.19               | 76.24               |
| 27                                      | stb-closed-loop | GAIN         | 42.34   |      |        |           | 37.35  | 45.39  | 45.39               | 37.35               |
| 27                                      | stb-closed-loop | GBW          | 76.1M   |      |        |           | 73.62M | 78.47M | 73.62M              | 78.47M              |
| Parameters: VDD=990m, temperature=42.5  |                 |              |         |      |        |           |        |        |                     |                     |
| 28                                      | stb-closed-loop | Gain Margin  | 24.32   |      |        |           | 23.67  | 24.56  | 24.56               | 23.67               |
| 28                                      | stb-closed-loop | Phase Margin | 83.59   |      |        |           | 83     | 83.59  | 83.56               | 83                  |
| 28                                      | stb-closed-loop | GAIN         | 41.63   |      |        |           | 36.8   | 44.54  | 44.54               | 36.8                |
| 28                                      | stb-closed-loop | GBW          | 64.1M   |      |        |           | 61.58M | 66.92M | 61.58M              | 66.92M              |
| Parameters: VDD=990m, temperature=83.75 |                 |              |         |      |        |           |        |        |                     |                     |
| 29                                      | stb-closed-loop | Gain Margin  | 25.37   |      |        |           | 24.63  | 25.66  | 25.66               | 24.63               |
| 29                                      | stb-closed-loop | Phase Margin | 90.41   |      |        |           | 89.54  | 90.55  | 90.55               | 89.54               |
| 29                                      | stb-closed-loop | GAIN         | 40.9    |      |        |           | 36.23  | 43.65  | 43.65               | 36.23               |
| 29                                      | stb-closed-loop | GBW          | 53.9M   |      |        |           | 51.42M | 56.99M | 51.42M              | 56.99M              |
| Parameters: VDD=990m, temperature=125   |                 |              |         |      |        |           |        |        |                     |                     |
| 30                                      | stb-closed-loop | Gain Margin  | 26.35   |      |        |           | 25.54  | 26.7   | 26.7                | 25.54               |
| 30                                      | stb-closed-loop | Phase Margin | 96.86   |      |        |           | 95.81  | 97.14  | 97.14               | 95.81               |
| 30                                      | stb-closed-loop | GAIN         | 40.11   |      |        |           | 35.64  | 42.69  | 42.69               | 35.64               |
| 30                                      | stb-closed-loop | GBW          | 45.21M  |      |        |           | 42.79M | 48.41M | 42.79M              | 48.41M              |

Comparison between required specs and simulated specs:

| Parameter            | Simulated or obtained specs | Required Specs     |
|----------------------|-----------------------------|--------------------|
| Input referred Noise | 61.04 u                     | 30 u(not obtained) |
| CL                   | 20p                         | 20p                |
| RL                   | 10k                         | 10k                |
| PM                   | 83.76                       | >45 DEGREE         |
| GBW                  | 63.47MHz                    | >2MHz              |
| OUTPUT SWING         | 1.2764V                     | >1V                |
| GAIN(AV0)            | 41.737dB                    | >40db              |
| VDD-VSS              | 1.8                         | 1.8                |

6. Comment on your learning from this lab:



- The need for iterative transistor sizing to meet all op-amp specifications.
- Use of simulations to verify performance metrics like DC gain, GBW, phase margin, output swing, and noise.
- Importance of running Monte Carlo simulations to assess variation and ensure robustness across best and worst-case corners.
- How supply and temperature variations impact circuit performance.
- Comparing calculated vs. simulated results helps identify practical adjustments in design.
- Real-world analog design challenges, especially achieving low input referred noise.
- The combination of analytical and simulation approaches is essential for effective op-amp characterization and design